College Pricing in the U.S.

Emily Cook & Sarah Turner
Higher Education

Introduction

In the U.S. higher education market, posted tuition (hereafter, "tuition") and net tuition (tuition minus grant aid) are rarely identical, and neither is equal to the expenditures required to produce a college education. Students are often eligible for grant aid based on financial need or academic merit, with most aid awards being provided by governments and colleges themselves. Grant aid is one dimension along which colleges compete for students in a market with a mix of for-profit, nonprofit and public institutions. However, even students who pay full tuition often do not pay the full per-student cost of their education, as expenditures per student typically exceed tuition. This is possible because colleges receive funding from private and public sources, including direct appropriations from federal, state and local governments. The amount of these subsidies varies greatly across institutions.

A common public refrain is, "Why does college cost so much?" Our response begins by showing that the prices that students pay—and how they have changed over time—differ markedly by type of institution and student characteristics. Measuring net tuition trends is particularly insightful, as the recent movement toward "high tuition, high aid" models at some institutions has increased the variance in tuition among institutions and between students within institutions. After presenting the empirical facts, we summarize the state of the art of research on the causal factors underlying trends in college pricing. We also briefly summarize the research exploring the consequences of pricing trends and the impact of interventions intended to improve college access and reduce information barriers.

Key findings

Key finding #1: *Posted tuition has increased dramatically over the last quarter century, but net tuition has increased much more slowly than posted tuition has.*

Posted tuition is what colleges advertise on their websites. From 1987 to 2020, (in-state) posted tuition increased by 203% at public four-year colleges and by 136% at private four-year nonprofit colleges. In the most recent decade, the rate of growth slowed at public institutions, and since 2020, tuition levels have declined in real terms. Net tuition is often lower than posted tuition, as it is the price paid by students after scholarships and grant aid are applied. While (in-state) tuition increased by 94% at public four-year colleges and by 54% at private four-year nonprofit colleges between 2000 and 2021, (in-state) net tuition increased by just 2.6% at public four-year colleges and by 10.6% at private colleges.[1]

Key finding #2: *The net tuition paid by students from lower-income families has increased much more slowly over time than that paid by students from higher-income families.*

This difference in price across family income groups reflects increased government and institutional grant aid alongside increasing tuition.

Key finding #3: *Many factors may be driving rising tuition, including rising expenditures (supply-side causes) and changes in student demand (demand- side causes).*

To quantify the role of each factor in generating recent patterns in tuition and net tuition, more research is needed.

Key finding #4: *In theory, improving access to information about net tuition could overcome barriers to enrollment, especially for low-income students.*

However, in practice, successful strategies for improving information about net tuition are few and far between.

Evidence

Key finding #1: *Posted tuition has increased dramatically over the last quarter century, but net tuition has increased much more slowly than posted tuition has.*

Figure 1 documents the dramatic increase in inflation-adjusted tuition at different categories of public and private institutions. Between the 1987–1988 academic year and the peak of tuition levels in 2020–2021, tuition rose at private R1 (highest research activity) institutions by 124%, and in-state tuition at public R1 institutions rose by 213%.[2] The percentage increases in tuition for other college types were similar to those at R1 institutions, with the exception of public community college tuition, which increased by 118% over the same period. In the two years following the COVID- 19 pandemic, tuition did not keep up with steep inflation, yielding the first downward movement in inflation-adjusted tuition in the last 35 years.

Tuition increases capture headlines about college affordability. However, net tuition—defined as tuition minus grants and aid that does not need to be repaid—better captures what students pay for college tuition. Institution-level data on average grants from all sources are available starting in the 2000–2001 academic year. Since that year, (in-state) tuition increased by 94% at public four-year colleges and by 54% at private four-year nonprofit colleges. During the same period, (in-state) *net tuition* increased by just 2.6% at public four-year colleges and by 10.6% at private colleges.

Key finding #2: *The net tuition paid by students from lower-income families has increased much more slowly over time than that paid by students from higher-income families.*

Trends in net tuition vary across colleges and income groups. Our discussion of trends in net tuition across income groups is based on our previous work on this topic.[3] In Figure 2, Panel A shows net tuition and fees by income group by college type for public institutions, while Panel B shows the same for private institutions. Only students who received some form of federal aid (either grants or loans) are included in the data by income group. Most other students pay full tuition. In these figures, we see increased price discrimination at both public and private institutions. That is, different students pay different prices for the

same college experience, and the magnitude of these price differences has increased over time. At public R1 institutions in 2021, students in the $48–$75K family income group paid approximately $4,000 less in net tuition than they did in 2008, in constant 2022 U.S. dollars. For private R1 institutions, students in this income group paid $13,700 less relative to 2008.[4] This trend of widening price dispersion across low-and middle-income families is common across public and private institutions. One distinction between private and public college pricing patterns is that at private colleges, students from higher-income families (e.g., the $110K+ group) who qualified for federal aid, including loans, were more likely to see price declines.[5]

The tuition discounts that generate price discrimination in higher education come in the form of scholarships or grants from a variety of sources, including state and federal governments as well as institutional and private sources. In recent decades, total grant aid has grown both in levels and as a share of aid to undergraduates. A recent report from the College Board shows that grant aid composed 48% of all aid to undergraduates in 2002–2003 but has grown to account for 64% of all aid as of 2022–2023.[6] From the same report, we learn that institutional aid (grant aid provided by colleges) has been one of the largest sources of growth in grant aid over the last two decades. Between the 2002–2003 and 2022–2023 academic years, institutional grants grew as a share of total aid sources (including loans) from 18% to 34%, while government grant aid remained relatively steady as a *share* of all aid. These raw statistics suggest that the recent increase in price discrimination across income groups can be largely attributed to increased grant aid from institutions themselves, rather than to state or federal grant aid.

Thus far, we have presented the basic trends: inflation-adjusted tuition had been rising until very recently, but aid increased simultaneously. Hence, the prices paid by students at different institutions and from different family income circumstances can differ widely. In the remainder of this article, we consider the possible determinants of these trends, the implications of these pricing trends for students, and the potential effects of interventions such as increased grant aid and reduced information barriers.

Key finding #3: *Many factors may be driving rising tuition, including rising expenditures (supply-side causes) and changes in student demand (demand- side causes).*

The increase in posted college tuition documented in the previous section has spurred research investigating its causes. One set of hypotheses suggests that tuition has increased because expenditures per student have increased. Increases in expenditures are hypothesized to stem from inefficiencies, administrative bloat or the phenomenon referred to as "Baumol's cost disease." Another theory known as the "revenue theory of costs" suggests that expenditures increase whenever revenue increases, but it is not as clear that this theory would explain increases in tuition alongside increases in expenditures. We discuss hypotheses regarding increased expenditures in the next subsection. Alternative explanations for increased tuition focus on the demand side, arguing that demand for a college education has increased over time. Increased demand could be driven by high wage returns to college completion and/or the increased availability of loan and grant funding. The suggestion that loan and grant aid may generate increased tuition is commonly referred to as the Bennett Hypothesis.[7] We discuss this hypothesis and other demand-side theories. Another set of explanations for increased tuition focuses on the unique role of state government policies for public universities, where state government policies such as reductions in state appropriations and a relaxation of legislative control might drive increases in tuition at public universities, at

least in some states.[8] These explanations are the focus of the last main subsection. Most of these hypotheses and the associated research do not address the *combination* of increasing tuition and increasing institutional grant aid, instead focusing on posted tuition.

Explanations based on rising expenditures

Per-student expenditures have been on the rise alongside tuition. Education and general expenditures rose from $33,511 per student in 1987 to $48,898 per student in 2021–2022,[Std. Err.1] on average, at private four-year universities. At public four-year institutions, the corresponding statistics are $24,467 in 1987 and $37,655 as of the latest data from the 2021–2022 academic year. The simultaneous increase in expenditures and tuition suggests a potential connection between the two, prompting research into the underlying causes of rising expenditure. Some researchers and commentators have pointed to increasing administrative layers and inefficiencies, although (perhaps due to data limitations) there is not much recent work investigating this hypothesis.[9]

Another explanation for rising expenditures—Baumol's cost disease—posits that in labor-intensive industries such as higher education, wages rise as a result of productivity gains in other sectors and the accompanying pressure on the labor market; however, productivity in education does not increase at the same rate.[10] Consequently, colleges must raise wages to attract and retain qualified faculty and staff, leading to higher expenditures. Since the output of educational services (such as teaching and mentoring students) is difficult to automate and requires significant human input, these increases in wage rates are not offset by technology-driven productivity gains in education.

Empirical analysis of the cost disease channel includes one study that compares prices in higher education to prices in other industries characterized by a highly educated workforce and little ability to benefit from technology-driven productivity gains (including physicians, doctors, and legal professionals).[11] This study concludes that the cost disease is likely an important driver of price increases in higher education. Another approach, which simultaneously models the actions of both colleges and students, also finds support for the cost disease argument, showing that a model with productivity growth in the noneducation sector and skill-biased technical change can match empirical trends in expenditures.[12] Although existing work suggests an important role for the cost disease channel, more work is needed to quantify the "pass-through" of rising expenditures to posted tuition increases and to assess the importance of the cost disease explanation compared to other factors affecting tuition increases.

Another explanation for changes in the structure of tuition identifies increased stratification in the market for higher education resulting from greater market integration and growing private subsidies among top institutions as a driver of increased expenditures per student. To the extent that the members of an elite group of institutions compete along the dimension of quality rather than price in a national market, added resources are used to attract stronger students and to spend more per student. Indeed, when these institutions face increases in demand, they generally do not respond with increases in capacity; instead, they increase selectivity.[13]

This behavior is often aligned with the revenue theory of costs, which posits that institutions' expenditures are driven by their available revenue.[14] Empirical analysis of this theory is limited. One study points out that the revenue theory of costs is specific to the

higher education market (e.g. because colleges are quality maximizing), while the cost disease theory applies to a great number of labor-intensive industries. The authors of the study use this distinction between the two theories to argue that the cost disease is a more likely explanation, as certain cost measures have trended similarly in higher education and comparable industries.[15]

Identifying and quantifying the factors underlying rising higher education expenditures constitute a crucial area for future research, as the policy implications of different proposed causes can vary. Moreover, it is important to determine the extent to which increased expenditures are transferred to students through grant and institutional aid. If the revenue theory of costs holds, then subsidies increase expenditures—but may also improve quality. On the other hand, if rising input prices are the primary explanation for increasing expenditures, then subsidizing higher education should reduce effective costs, which may pass through to students in the form of reduced tuition.

Demand-side explanations

Increasing demand for college could cause increased tuition by allowing colleges to raise tuition without losing enrollment or decreasing selectivity. This increased demand could be fueled by changes in demographics or high labor market returns to college. Alternatively, demand could be stimulated by the increased availability of loan and grant funding. Another demand-side explanation for rising tuition points to rising income inequality and differences in students' willingness to pay across income levels.[16] Below, we focus our discussion on two potential demand drivers: the labor market and increased aid.

It is likely that increases in the college wage premium may induce higher demand for college enrollment, as the evidence shows that students are sensitive to current labor market conditions and expectations of future earnings. Some work uses variation in labor market conditions over time or across geography to show that demand for college (and for specific programs) is sensitive to labor market conditions.[17] There is also a vast literature demonstrating a positive relationship between expected wage returns and the choice to enroll.[18]

Another potential driver of increased demand for college is the increased availability of loan and grant aid, a theory often referred to as the Bennett Hypothesis.[19] According to this theory, when the government provides more financial aid, colleges respond by raising their tuition rates, knowing that students can afford to pay more due to the increased aid. This response by colleges creates a cycle where financial aid, intended to make college more affordable, instead leads to higher tuition costs, reducing the intended benefit of the aid. It is difficult to test this hypothesis because there are no clear treatment and control institutions for any large aid policy. Institutions in the U.S. are connected via competition in a national market; hence, even comparing across states with different aid policies does not provide an unaffected control group. Evidence using quasi-experimental methods in traditional nonprofit private and public sectors is mixed, depending on the context.[20] Studies focusing on the for-profit sector have found compelling evidence supporting the Bennett Hypothesis. Studies demonstrate that changes in federal student aid in the context of Pell Grants and the GI Bill led to significant tuition responses among for-profit institutions.[21] Importantly, however, these findings may not generalize to other sectors of higher education, where institutions have somewhat different objectives, types of students receiving federal financial aid, and other sources of revenue.

Analysis of the Bennett Hypothesis might be best understood by modelling the responses of students and colleges (using what economists call a "structural model"). This approach seeks to estimate the preferences and objectives governing the choices of students and colleges and to use these estimates to simulate the effects of policies and other exogenous changes to the market (e.g., demographic changes). However, this approach can also be challenging to implement due to the unusual features of this market, including multiple sources of financial aid, imperfect information, imperfection competition, and a decentralized admissions process. A recent structural model of the U.S. higher education market infers college objectives and simulates equilibrium pricing and admission responses to increased grant aid at the state level.[22] The findings support the Bennett Hypothesis, with some nuance. In these simulations, government grant aid generates increased tuition alongside higher aid and increased admissions standards at the colleges that are targeted by the aid program.

There are several other structural models of college competition in the U.S.[23] Most of these studies demonstrate that price discrimination (charging different prices to different students) is an empirically important feature of this market, although most do not include a direct test of the Bennett Hypothesis. Structural models that include a test of the Bennett Hypothesis find support for it.

State governments and public university tuition

State governments are heavily involved in the provision of higher education in their states through appropriations to public universities and direct legislative control over some higher education policies. In some cases, state governing bodies have the authority to dictate tuition levels or cap tuition increases. Researchers have used variation in these policies to investigate the effects on the prices paid by students.

Research has demonstrated a causal link between state appropriations and tuition through the use of instrumental variables based on variation in state budgets.[24] Estimates from these studies suggest a pass-through of appropriations to tuition levels of approximately 30–45%, depending on the type of institution. In other words, increasing appropriations by $100 per student would decrease tuition by approximately $30–$45. However, our previous work shows that while appropriations trended upward during the period between the Great Recession and the COVID-19 pandemic—which would imply reductions in tuition levels—tuition continued to rise. This finding suggests that changes in state appropriations are not a complete explanation for recent trends in price discrimination and tuition.

State governments have policy levers other than appropriations that can also affect tuition at public institutions. A recent study in Texas showed that when the state government relinquished control of tuition-setting authority and allowed public colleges to set their own tuition levels, tuition increased. However, aid to students also increased, resulting in improved outcomes for some low-income students.[25]

Most state tuition regulatory policies target reducing tuition or, at least, capping increases. When states impose tuition caps or freezes, they are typically effective in reducing tuition at least for a short time. However, research suggests that colleges have responded to tuition freezes in many cases by decreasing student financial aid or by implementing large tuition increases after the temporary cap or freeze was lifted.[26]

Summary

Explanations for changing tuition levels are complex and involve many factors, and it is likely that the importance of each factor has changed over time and varies by type of institution. Macroeconomic forces that impact input prices may dominate in some periods but not others. Moreover, differences in institutional mission—such as those between broad-access institutions and research universities—impact the extent to which institutions are likely to award substantial institutional aid, which widens the gap between the posted price and the net prices paid by different groups of students. The divergence evident in the most recent period between posted tuition and the net tuition for low- and middle-income[Std. Err.2] families at research universities serves as an important illustration of this point.

Identifying the causes of increased tuition is policy relevant. For example, if demand-side subsidies have the unintended consequence of increasing tuition, then students (particularly low-income students) may not reap the full intended benefits of increases in grant aid. On the expenditure side, the revenue theory of costs suggests that subsidies can increase the cost of providing college education, while the cost disease theory suggests that subsidies help alleviate the consequences of technological progress for higher education costs. Some new work aims to quantify the role of supply- and demand-side factors in generating these trends, suggesting a large role for the cost disease and demand-side explanations.[27] These models provide an ambitious starting point. In the future, continued improvements in data availability and econometric methods may enable increasingly sophisticated models of higher education to help understand the reasons for rising tuition.

Key finding #4: *In theory, improving access to information about net tuition could overcome barriers to enrollment, especially for low-income students*.

Increasing tuition may deter prospective students, particularly those from low- and middle- income families, from enrolling in college due to financial barriers. Economic theory is unequivocal on this point: the quantity demanded almost always falls when price rises. Hence, increasing what students must pay is expected to lead to a reduction in enrollment, with these effects being magnified for those from lower-income families. With full information, theory would predict that the effect would be the same for increases in tuition and decreases in financial aid, leaving net tuition as the relevant variable for understanding price responsiveness. However, perceptions of affordability may not track net tuition, especially given that the net tuition is not known until after students have already been forced to make important college decisions.

Enrollment responses to changes in tuition

There is a long history of attempts to estimate the sensitivity of demand to the price of college (what economists call the "price elasticity of demand"). For example, a 1988 survey summarizes a wide range of correlational studies of the link between tuition and enrollment, showing considerable variation in estimated effects.[28] Estimation strategies that use variation in tuition across colleges within states to estimate the price elasticity of demand often produced estimates indicating a near-zero effect.[29] However, the empirical validity of many of these panel data approaches has been called into question as researchers have focused on the nature of variation producing changes in tuition. Enrollment decisions are often sensitive to the same factors that move tuition levels: for

example, upward pressure on tuition from a reduction in state support during weak economies may be correlated with individuals' increased likelihood of enrollment during weak local labor markets.

Using contemporary econometric tools, a small number of studies take advantage of the variation in community college tuition across localities to estimate enrollment responses. For example, one study in Texas shows that access to discounted (in-district) community college tuition has a substantial positive impact on enrollment.[30] Another study that uses Michigan data employs a boundary fixed effects strategy and shows that reductions in community college tuition levels not only increase the total number of enrollees but also shift enrollment from for-profit institutions and other vocationally oriented programs.[31]

Overall, causal estimates of the impact of changes in posted tuition on enrollment are hard to obtain for several reasons. First, there is obviously no opportunity for "random assignment" or a clear experimental design. Second, available quasi-experimental estimates are necessarily local, applying only to particular types of colleges or student populations. Third, tuition responsiveness likely depends on the availability of close substitutes, with concurrent tuition adjustment at many institutions in a market having a smaller impact than a tuition change for a single institution. Finally, to the extent that institutions offer financial aid, posted tuition changes have different impacts on net tuition for different groups of students, depending on eligibility for aid.

Enrollment responses to changes in grant aid

While reductions in tuition and increases in grant aid have identical mathematical effects on net tuition, enrollment responses may not be the same due to students' misperceptions of aid availability, the cost of applying for grants (including paperwork and documentation requirements), and uncertainty about whether promised aid will materialize when students need it. Studies on the direct impact of grant aid on enrollment have yielded a wide range of estimates depending on the design of the aid programs. Recall that grant aid comes from multiple sources with different criteria: federal Pell Grants are means tested, with eligibility determined by a formula based on data submitted on the Free Application for Federal Student Aid (FAFSA) form; states provide grants based on need and merit criteria; and some institutions provide additional need/merit aid.

Dating back to 1972, the federal Pell Grant program provides a means-tested portable grant to low-income students for enrollment in post-secondary education. With a maximum value of $7,395 in 2023–2024, the expectation is that Pell Grants would lower the net price for eligible students and increase college enrollment rates. However, since the inception of the program, the empirical evidence has been more nuanced. Several studies from the 1980s and 1990s found that students from low-income families who would have been Pell eligible experienced little change in their college enrollment rates after 1973.[32] Nonetheless, the results for independent students who are generally older than age 24 show more responsiveness.[33] A more extensive discussion of the FAFSA and federal grant aid is available.

In the discussion of the limited evidence of enrollment effects of the Pell Grant program, two types of explanations predominate. The first, which has some empirical support, is that institutions effectively "crowd out" additional federal aid for low-income students by replacing institutional aid with federal grants while also increasing aid somewhat above the Pell eligibility thresholds.[34] The second explanation, which we address in greater detail

in the next section, is that the onerous process of applying for financial aid may limit the extent to which students understand eligibility for Pell Grants and net tuition at the stage when they are making their college choices.

The limited enrollment response to the Pell Grant program does not generalize to all federal grant aid programs; the Social Security Student Benefit (SSSB) program and the World War II-era GI Bill stand as notable counterexamples with substantial enrollment responsiveness. What distinguishes these programs in their implementation is the clarity of their eligibility criteria and the relative simplicity of application.[35]

State merit aid programs are another form of aid that provides reductions (in some cases, 100%) in tuition for students who meet particular academic requirements, such as GPA or test score thresholds. What these programs share with the SSSB and GI Bill is transparency; however, they differ in their "portability," as they are generally limited to enrollment at public, in-state options. One of the earliest studies on this form of aid was an evaluation of the Georgia Hope program that used a difference-in-differences approach. The study found substantial enrollment effects.[36] Another study examined the West Virginia PROMISE scholarship, finding enrollment effects and sizeable impacts on credit attainment and BA completion.[37] Merit aid programs have been studied extensively, and more details are available in <HYPERLINK TO SCHOLARSHIPS AND PROMISE PROGRAMS>.

Even as categorical programs and state merit aid programs appear to impact enrollment, there is less evidence that the workhorse tools, namely, federal Pell Grants and institutional aid, impact enrollment. It may be the case that students (and their families) are not well informed about the availability of grant aid and the associated net price of enrollment at different institutions.[38]

Information interventions and other reforms

If students and their families are not aware of the availability of financial aid, then they may perceive high-quality, high-tuition options as unaffordable, even when these institutions may actually charge a lower net price than would a lower-resourced college with low posted tuition. As a result, students may lose out on high-return options or forgo college altogether.

For students who are likely eligible for federal financial aid along with need-based aid offered by institutions, one central challenge is that the net price may be hard to discern prior to application. While simplification of needs assessment may provide some opportunity to "estimate" the federal formula that determines Pell Grants, students must still wait until after acceptance to know the level of institutional aid and aid from other sources forming their net price. The passage of the Higher Education Opportunity Act of 2008 required colleges to post net price calculators. However, utilization of these tools appears to be limited, and researchers have also called into question their accuracy, given out-of-date data and other problems.[39]

A related friction in the market is the complexity of the aid application process, including the completion of the FAFSA form, which has been described as overly complex and a potential deterrent to application.[40] In a particularly innovative study, a group of researchers partnered with a tax preparation provider to help students (and families) complete the FAFSA form simultaneously with income tax preparation, as much of the

same information is required for both forms.[41] Those students receiving the intensive treatment that included assistance with filing the FAFSA form were more likely to matriculate at a post-secondary institution in a subsequent year.

Given concerns that students do not enroll or do not consider a full range of options because they are unaware of the availability of financial aid, recent policy interventions aim to make net tuition (or net price) transparent to low- and middle-income students. One set of interventions targets high-achieving, low-income students who often forgo opportunities at selective institutions at which net tuition would likely be lower and educational resources are greater than those at the institutions that they choose to attend.

In one study, researchers used mail-based outreach to provide information showing that the divergence between tuition and net tuition is particularly large at research-intensive private colleges and state flagship institutions.[42] Provision of this information, along with other supporting materials to facilitate application, had a large impact on application in the treatment group of the experiment. In another study, researchers focused on the context of the University of Michigan where they targeted outreach to high-school students who were almost certainly aid eligible and likely qualified for admission.[43] They found that students in the treatment arm of the experiment were more likely not only to apply to the University of Michigan but also to be admitted and matriculate. For this group of students, greater information about net price had a larger impact on where these students chose to enroll than on whether they chose to enroll.

Conclusion

In standard economic models, individual choices about whether to attend college and where to attend respond to net tuition (the difference between tuition and grant aid). The determination of net tuition necessarily derives from variables affecting tuition and grant aid, with some of these factors being determined by institutions and others being determined by state and federal policy. However, the extent to which individuals respond to net tuition rather than posted tuition depends heavily on full information about net tuition.

The empirical evidence on changes over time in post-secondary pricing demonstrates the complexity of the higher education market. While broadly rising tuition levels (with qualification for the last several years) is a common refrain, changes in the net tuition faced by students are context specific, depending on a student's characteristics (particularly family income) and the type of institution under consideration. The most resource-intensive institutions, including research universities in both the private and public sectors as well as private liberal arts colleges, demonstrate increased variation in net prices. The most affluent students face rising net tuition, while low- and middle-income students face some declines.

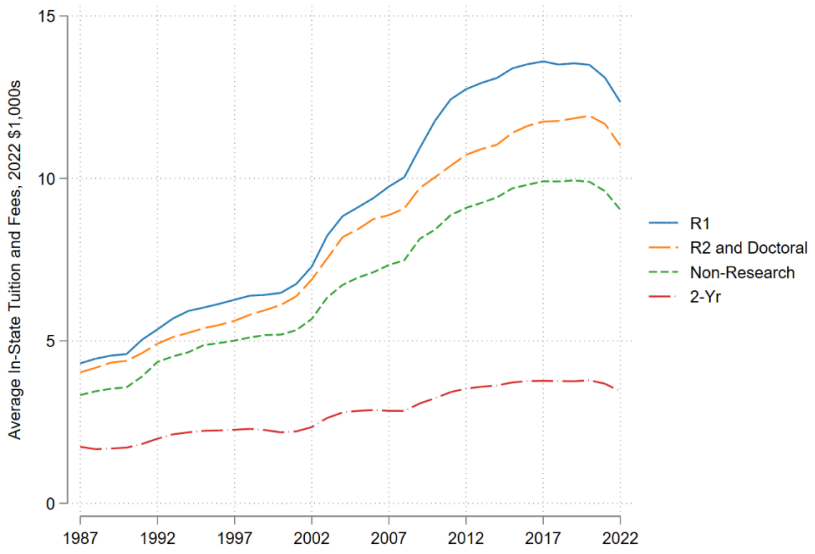
A considerable amount of ink has been spilled by pundits and public officials excoriating colleges and universities for rising tuition levels, while economists and other analysts have attempted to unpack the causes of tuition increases. While some critiques of the behavior of colleges and universities are certainly on point, the exclusive focus on posted tuition levels is increasingly irrelevant in an environment in which there is a large divergence between tuition levels and net tuition for many students. As we showed in key finding #2, net tuition has not actually been rising the way that alarming headlines suggest.

While, in theory, net tuition is the price that should affect individuals' decision-making, there are good reasons to believe that this variable is not well understood by students and their families. Frictions tied to both the complexity of financial aid application and the lack of "point of application" information about aid availability may limit individuals' knowledge about net tuition and, hence, the extent to which it impacts decision-making.

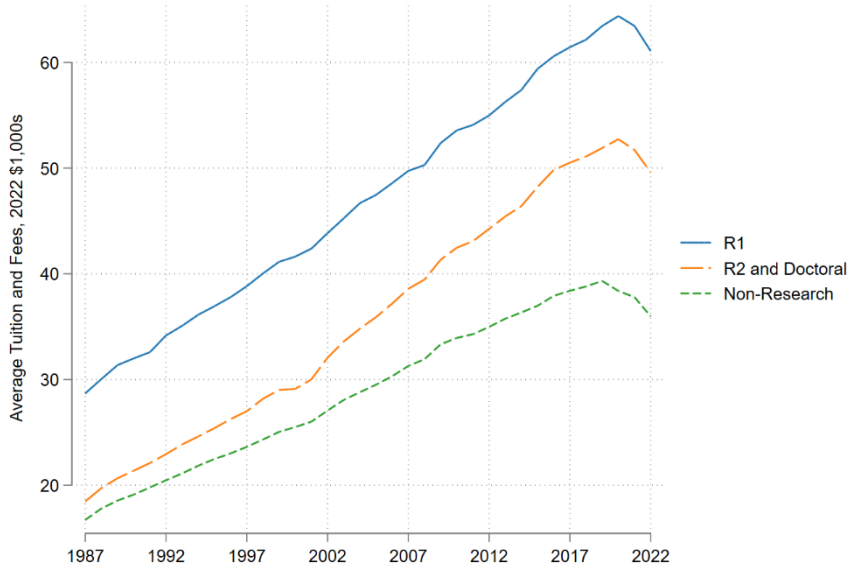
Finally, it is important to emphasize the heterogeneity in quality and the differentiation of products in higher education. Hence, any tuition level (or net tuition faced by an individual) needs to be interpreted in the context of an individual's expected return. Low or zero tuition may not serve individuals or society if the education provided does not yield a sufficiently high-quality education or high education returns.

Figure 1: Tuition over Time by Institution Type

Public Colleges



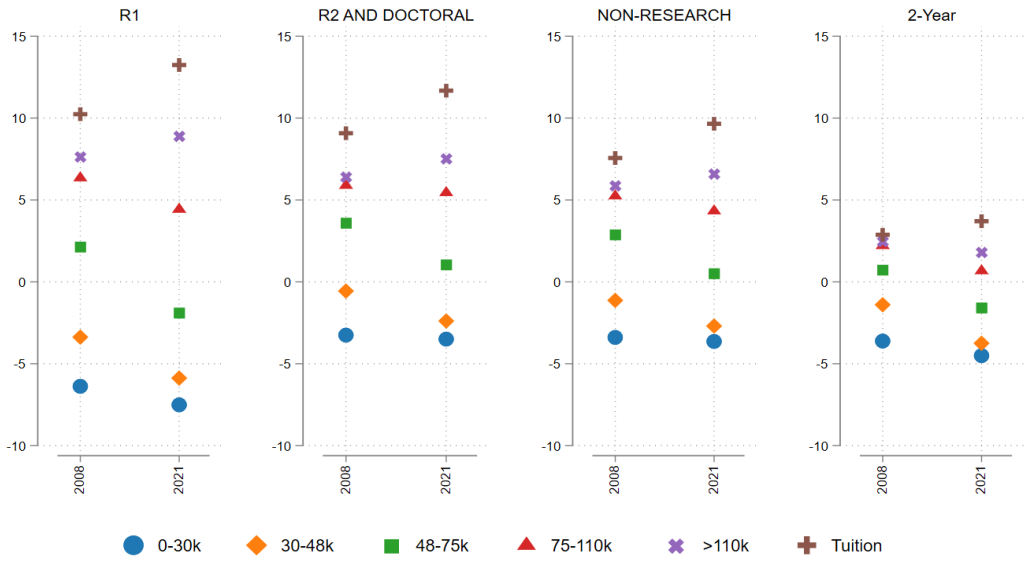
Private Colleges



Notes: Authors' calculations from the Integrated Postsecondary Education Data System (IPEDS). Tuition is the posted in-state tuition for public institutions and the posted tuition for private institutions. The definitions of R1, R2 and Doctoral, and Non-Research institutions are based on the 2005 Carnegie Classification codes. The averages are weighted by the number of full-time equivalent undergraduate students.

Figure 2: Net Tuition by Student Income and College Type over Time

Public Colleges



Private Colleges

Notes: Authors' calculations from the Integrated Postsecondary Education Data System (IPEDS). Net tuition is posted tuition minus grants and aid that does not have to be repaid from all sources. IPEDS began collecting net price data by income group in 2008. The definitions of R1, R2 and Doctoral, and Non-Research institutions are based on the 2005 Carnegie Classification codes. The averages are weighted by the number of full-time equivalent undergraduate students.

Endnotes and references

[1] These statistics and those under key finding #1 are from the authors' calculations based on enrollment-weighted average tuition and federal, state, and institutional grant aid from the Integrated Postsecondary Education Data System (IPEDS).
[2] We begin our analysis in 1987, which is the first year of our data series from the IPEDS of the National Center for Education Statistics.
[3] Cook, E. E., and S. Turner. 2022. Progressivity of Pricing at US Public Universities. Economics of Education Review 88.
[4] It is also true that inflation has changed where these income groups—which are measured in nominal dollars—lie in the income distribution. However, inflation for the period discussed here was relatively low with the exception of 2021.
[5] The same pattern holds for the net cost of attendance, which is tuition, fees, room, and board net of grant aid. A figure showing these data is available upon request from the authors.
[6] College Board Research, J. Ma, S. Baum, M. Pender, and D. Bell. 2023. Trends in College Pricing 2023.
[7] Bennett, W. J. 1987, February. Our Greedy Colleges. The New York Times. February 18. https://www.nytimes.com/1987/02/18/opinion/our-greedy-colleges.html.
[8] Webber, D. A. 2017. State Divestment and Tuition at Public Institutions. Economics of Education Review 60: 1–4; Andrews, R. J., and K. M. Stange. 2019. Price Regulation, Price Discrimination, and Equality of Opportunity in Higher Education: Evidence from Texas. American Economic Journal: Economic Policy 11(4): 31–65.
[9] Bergmann, B. R. 1991. Bloated Administration, Blighted Campuses. Academe 77(6): 12–16; Leslie, L. L., and G. Rhoades. 1995. Rising Administrative Costs: Seeking Explanations. The Journal of Higher Education 66(2): 187–212; Rhoades, G. 1998. Reviewing and Rethinking Administrative Costs. Higher Education: Handbook of Theory and Research. 111–147; Hedrick, D. W., C. S. Wassell, and S. E. Henson. 2009. Administrative Costs in Higher Education: How Fast Are They Really Growing? Education Economics 17(1): 123–137.
[10] Baumol, W. J., and W. G. Bowen. 1966. Performing Arts: The Economic Dilemma; a Study of Problems Common to Theater, Opera, Music, and Dance. The Twentieth Century Fund; Bowen, W. G. 2012. The "Cost Disease" in Higher Education: Is Technology the Answer? The Tanner Lectures. Stanford University.
[11] Archibald, R. B., and D. H. Feldman. 2018. Drivers of the Rising Price of a College Education. Midwestern Higher Education Compact. August.
[12] Jones, J. B., and F. Yang. 2016. Skill-Biased Technical Change and the Cost of Higher Education. Journal of Labor Economics 34(3): 621–662.
[13] Hoxby, C. M. 2009. The Changing Selectivity of American Colleges. Journal of Economic Perspectives 23(4): 95–118; Bound, J., and S. Turner. 2007. Cohort Crowding: How Resources Affect Collegiate Attainment. Journal of Public Economics 91(5–6): 877–899.
[14] Ehrenberg, R. G. 2002. Tuition Rising: Why College Costs So Much. Harvard University Press.
[15] Archibald, R. B., and D. H. Feldman, D. H. 2008. Explaining Increases in Higher Education Costs. Journal of Higher Education 79(3): 268–295.
[16] Cai, Z., and J. Heathcote. 2022. College Tuition and Income Inequality. American Economic Review 112(1): 81–121.
[17] Betts, J. R., and L. L. Mcfarland. 1995. Safe Port in a Storm: The Impact of Labor Market Conditions on Community College Enrollments (Tech. rep. No. 4). Autumn; Barr, A., and S. Turner. 2015. Out of Work and into School: Labor Market Policies and College Enrollment during the Great Recession. Journal of Public Economics 124: 63–73; Charles, K. K., E. Hurst, and M. J. Notowidigdo. 2018. Housing Booms and Busts, Labor Market Opportunities, and College Attendance. American Economic Review 108(10): 2947–2994; Foote, A., and M. Grosz. 2020. The Effect of Local Labor Market Downturns on Postsecondary Enrollment and Program Choice. Education Finance and Policy 15(4): 593–622; Acton, R. K. 2021. Community College Program Choices in the Wake of Local Job Losses. Journal of Labor Economics 39(4): 1129–1154.
[18] Willis, R., and S. Rosen. 1979. Education and Self-Selection. Journal of Political Economy 87(5): 7–36; Taber, C. R. 2001. The Rising College Premium in the Eighties: Return to College or Return to Unobserved Ability? Review of Economic Studies 68(3): 665–691; Fang, H. 2006. Disentangling the College Wage Premium: Estimating a Model with Endogenous Education Choices. International Economic Review 47(4): 1151–1185; Dillon, E. W. 2017. The College Earnings Premium and Changes in College Enrollment: Testing Models of Expectation Formation. Labour Economics 49: 84–

94; Bleemer, Z., and B. Zafar. 2018. Intended College Attendance: Evidence from an Experiment on College Returns and Costs. Journal of Public Economics 157: 184–211.
[19] Bennett (1987).
[20] Long, B. T. 2004. How Do Financial Aid Policies Affect Colleges? The Institutional Impact of the Georgia HOPE Scholarship. Journal of Human Resources 39(4): 1045–1066; Long, B. T. 2006. College Tuition Pricing and Federal Financial Aid: Is there a Connection? (Tech. rep.); Singell, L. D., and J. A. Stone. 2007. For Whom the Pell Tolls: The Response of University Tuition to Federal Grants-in-Aid. Economics of Education Review 26(3): 285–295; Turner, N. 2012. Who Benefits from Student Aid? The Economic Incidence of Tax-based Federal Student Aid. Economics of Education Review 31(4): 463–481.
[21] Cellini, S. R., and C. Goldin. 2014. Does Federal Student Aid Raise Tuition? New Evidence on For-Profit Colleges. American Economic Journal: Economic Policy 6(4): 174–206; Baird, M., M. Crawford, T. Miller, and J. Wenger. 2022. Veteran Educators or For-Profiteers? Tuition Responses to Changes in the Post 9/11 GI Bill. Journal of Policy Analysis and Management 41(4): 1012–1039.
[22] Cook, E. 2024. Market Structure and College Access in the US; Cai and Heathcote (2022); Epple, D., R. Romano, and H. Sieg. 2006. Admission, Tuition, and Financial Aid Policies in the Market for Higher Education. Econometrica 74(4): 885–928; Fu, C. 2014. Equilibrium Tuition, Applications, Admissions, and Enrollment in the College Market. Journal of Political Economy 122(2): 225–281; Epple, D., R. Romano, S. Sarpça, H. Sieg, and M. Zaber. 2019. Market Power and Price Discrimination in the US Market for Higher Education. RAND Journal of Economics 50(1): 201–225; Gordon, G., and A. Hedlund. 2018. Accounting for the Rise in College Tuition. In Education, Skills, and Technical Change: Implications for Future US GDP Growth. Edited by C. R. Hulten and V. A. Ramey. University of Chicago Press. 357–401; Gordon, G., and A. Hedlund. 2022. Accounting for Tuition Increases across U.S. Colleges (Tech. rep.).
[23] Cai and Heathcote (2022).
[24] Cook and Turner (2022); Webber (2017).
[25] Andrews and Stange (2019). Increased grant aid was to some degree mandated by the deregulation policy in this case.
[26] Miller, L., and M. Park. 2022. Making College Affordable? The Impacts of Tuition Freezes and Caps. Economics of Education Review 89.
[27] Gordon and Hedlund (2018). Gordon and Hedlund (2022).
[28] Leslie, L., and P. Brinkman. 1988. The Economic Value of Higher Education. American Council on Education/Macmillan Series on Higher Education.
[29] Fitzpatrick, M., and S. Turner. 2008. Blurring the Boundary: Changes in Collegiate Participation and the Transition to Adulthood. In The Price of Independence: The Economics of Early Adulthood. Edited by S. Danziger and C. Rouse. Russell Sage Foundation. 107–138; Kane, T. 1995. Rising Public College Tuition and College Entry: How Well Do Public Subsidies Promote Access to College? National Bureau Of Economic Research.
[30] Denning, J. T. 2017. College on the Cheap: Consequences of Community College Tuition Reductions. American Economic Journal: Economic Policy 9(2): 155–188.
[31] Acton, R. 2021. Effects of Reduced Community College Tuition on College Choices and Degree Completion. Education Finance and Policy 16(3): 388–417.
[32] Kane (1995); Hansen, W. L. 1983. Impact of Student Financial Aid on Access. Proceedings of the Academy of Political Science 35(2): 84; Kane, T. J. 1994. College Entry by Blacks since 1970: The Role of College Costs, Family Background, and the Returns to Education. Journal of Political Economy.
[33] Seftor, N. S., and S. E. Turner. 2002. Back to School: Federal Student Aid Policy and Adult College Enrollment Back to School Federal Student Aid Policy and Adult College Enrollment. The Journal of Human Resources 37(2): 336–352.
[34] Turner, L. J. 2017. The Economic Incidence of Federal Student Grant Aid.
[35] Dynarski, S. M. 2003. Does Aid Matter? Measuring the Effect of Student Aid on College Attendance and Completion. American Economic Review 93(1): 279–288; Bound, J., and S. Turner. 2002. Going to War and Going to College: Did World War II and the G.I. Bill Increase Educational Attainment for Returning Veterans? Journal of Labor Economics 20(4): 784–815.
[36] Dynarski, S. 2000. Hope for Whom? Financial Aid for the Middle Class and Its Impact on College Attendance. National Tax Journal 53(3.2): 629–661.
[37] Scott-Clayton, J. 2011. On Money and Motivation: A Quasi-Experimental Analysis of Financial Incentives for College Achievement. The Journal of Human Resources 46(3).

[38] Hoxby, C., and S. Turner. 2013. Expanding College Opportunities for High-Achieving, Low-Income Students (Stanford Institute for Economic Policy Research Discussion Paper 12-014). 1–57; Avery, C., and T. J. Kane. 2004. Student Perceptions of College Opportunities: The Boston COACH Program. In College Choices: The Economics of Where to Go, When to Go, and How to Pay for It. University of Chicago Press. 355–394.
[39] Perna, L. W., J. Wright-Kim, and N. Jiang. 2019. Questioning the Calculations: Are Colleges Complying with Federal and Ethical Mandates for Providing Students with Estimated Costs? University of Pennsylvania Graduate School of Education.
[40] Dynarski, S. M., and J. E. Scott-Clayton. (2006). The Cost of Complexity in Federal Student Aid: Lessons from Optimal Tax Theory and Behavioral Economics. National Tax Journal 59(2): 319–356.
[41] Bettinger, E. P., B. T. Long, P. Oreopoulos, and L. Sanbonmatsu. 2012. The Role of Application Assistance and Information in College Decisions: Results from the H&R Block FAFSA Experiment. The Quarterly Journal of Economics. 1205–1242
[42] Hoxby and Turner (2013).
[43] Dynarski, S., C. J. Libassi, K. Michelmore, and S. Owen. 2021. Closing the Gap: The Effect of Reducing Complexity and Uncertainty in College Pricing on the Choices of Low- Income Students. American Economic Review 111(6): 1721–1756.

Suggested citation

Cook, Emily and Sarah Turner (2025). "College Pricing in the U.S.," *in Live Handbook of Research on Education Finance and Policy*, in Douglas Harris (ed.), Association for Education Finance and Policy, viewed MM/DD/YYYY, <WEB LINK>.